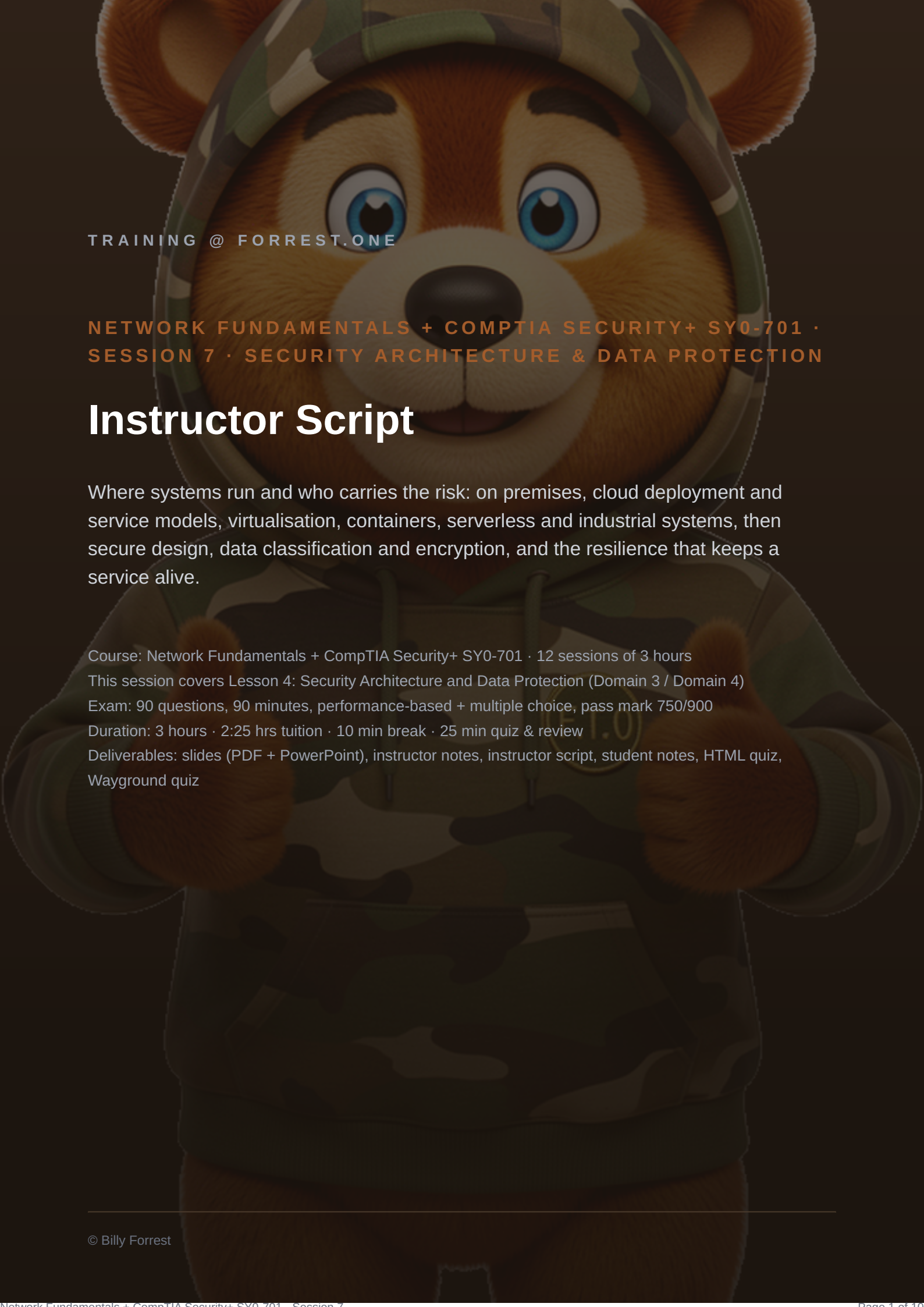




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 7 · SECURITY ARCHITECTURE & DATA PROTECTION**

Instructor Script

Where systems run and who carries the risk: on premises, cloud deployment and service models, virtualisation, containers, serverless and industrial systems, then secure design, data classification and encryption, and the resilience that keeps a service alive.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session covers Lesson 4: Security Architecture and Data Protection (Domain 3 / Domain 4)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Instructor Script — Security Architecture & Data Protection

A word-for-word delivery script for Session 7. Stage directions appear in square brackets. Each slide runs roughly 45 to 90 seconds of speech; activities and discussions expand the timings in the instructor notes.

Slide 1 · Title

Good day, and welcome to Session Seven. This is Security Architecture and Data Protection. We look at where systems run, how they are built, and how their data is defended in every state it occupies. Then we turn to resilience, keeping services alive when hardware fails and recovering data when it is lost. Every control we meet today either protects data or protects the systems that hold it. Architecture decisions are long lived, so a mistake here can persist for years.

Slide 2 · The shape of the session

Here is the shape of the session. First, architecture models: on premises, cloud deployment, cloud service models, virtualisation, containers, and the internet of things. Then secure design: infrastructure as code, secure network design, and how security controls are chosen. Then data protection: data types and states, classification, and encryption for data at rest and data in transit. Finally resilience: high availability, alternate sites, backup and recovery objectives, testing, and business continuity planning. Each chapter teaches a decision you will make in a real environment.

Slide 3 · Session objectives

By the end of these three hours you will be able to compare architecture models, place the responsibility line in each cloud model, secure compute platforms from the hypervisor up, reason about edge and industrial systems, design for containment, protect data by type and state, build in resilience, and plan for continuity. That is the whole session, and the quiz tests every part of it.

Slide 4 · On-Premises Architecture

We begin on premises, where the organisation owns the hardware and can touch it. The advantage is direct physical control, so server room access is regulated, recorded, and can require a biometric. The price is responsibility, because patching, maintenance, disaster recovery planning, and skilled staff all sit with you. Total cost of ownership covers not only purchase but power, cooling, and replacement. On premises still suits regulated industries, classified data, and workloads that need very low latency. Control and responsibility are the same coin.

Slide 5 · Cloud Deployment Models

Cloud changes the question to who shares the platform. A public cloud is shared among tenants and sold by providers such as Amazon Web Services, Microsoft Azure, and Google Cloud Platform. A private cloud is dedicated to one organisation and can sit on premises or with a hosting partner. A hybrid cloud combines both and lets data and applications move between them. A community cloud is shared by organisations with common needs, such as one industry standard. Notice that each model shifts the security duties between provider and customer.

Slide 6 · Cloud Service Models

Service models decide how far the provider reaches. Infrastructure as a Service gives you virtual machines, storage, and networks, and you secure everything above the hypervisor, including the operating system, patches, and applications. Platform as a Service manages the runtime, so your work is application security and data protection. Software as a Service runs the entire stack, leaving you to manage access and protect the data inside the application. The pattern is consistent: the more the provider manages, the less responsibility you carry, and the less control you hold.

Slide 7 · Activity: read the responsibility line

[Run the activity per the instructor notes. 6 minutes pairs, 4 minutes debrief.] Test the model with three quick cases. You rent a virtual machine and patch its operating system yourself, so that is infrastructure as a service. You deploy application code onto a managed runtime and never touch an operating system, so that is platform as a service. You use a hosted mail suite and manage only permissions and data, so that is software as a service. The line moves, and your job is to know where it sits.

Slide 8 · Compute and Edge Platforms

Chapter two moves inside the platform. Virtualisation, containers, and serverless computing each redraw the security boundary. Then we leave the data centre and meet devices that were never designed to be defended. And we finish with the industrial systems that keep power and water flowing. In every case the patch responsibility moves to somebody else, and that changes the risk as much as the technology does. Ask who applies the patches, and the architecture explains itself.

Slide 9 · Virtualisation Security

Virtualisation runs many virtual machines on one physical host, and everything depends on the hypervisor. A type one hypervisor runs directly on the hardware, so it has a smaller attack surface and is the more secure choice. A type two hypervisor runs on a host operating system, so a flaw in that host becomes a flaw in every guest. The attack to know is virtual machine escape, breaking out of a guest to reach the hypervisor. Sprawl is the quieter risk, because forgotten virtual machines are never patched. Remember that snapshots hold sensitive data and consume storage.

Slide 10 · Containerisation

Containers are a lighter form of virtualisation. Instead of a full operating system per instance, containers share the host kernel and package the application with its libraries into an image. That makes them quick to deploy and consistent everywhere, so Docker spread fast. The risk is the image, because a vulnerable library inside it is reproduced in every copy, so scan images and use trusted sources. Orchestration platforms such as Kubernetes add another layer to secure, including role based access control and network policy.

Slide 11 · Serverless Architecture

Serverless takes one more step away from the server. The provider runs the infrastructure, and your code is broken into functions that execute only when an event triggers them. Those functions are stateless and ephemeral, living just for the execution, which shrinks the attack surface. Amazon Lambda, Azure Functions, and Google Cloud Functions are the usual examples. The catch is the trigger itself, because an event is an entry point, so authenticate and validate everything that calls a function. And accept that you have less visibility inside the provider platform.

Slide 12 · Internet of Things Security

Now the internet of things: sensors, cameras, and embedded devices. The Internet of Things brings four risks worth naming. The devices are too small to run heavy security controls. They receive updates rarely, so known flaws stay open for a long time. They ship with default credentials that nobody changes. And there are so many of them that every device is another way in. The defences are segmentation, continuous monitoring, and lifecycle management through to decommissioning.

Slide 13 · Industrial Control Systems and SCADA

Industrial control systems run physical processes, so a breach can hurt people. Industrial control systems include distributed control systems, programmable logic controllers, and SCADA, which is Supervisory Control and Data Acquisition. A SCADA system has terminal units in the field, a human machine interface for operators, and a communications network joining them. Two things make these systems hard to secure. They use legacy protocols with no encryption or authentication, and they must stay available, because stopping a pump or a turbine can be dangerous.

Slide 14 · Discussion

[Run the discussion per the instructor notes.] Here is the convergence problem. Information technology protects confidentiality, while operational technology protects safety and availability, and the two cultures think differently. When the networks are joined for efficiency, an intruder who reaches the office network can reach the plant. The answer is not to keep them apart forever, but to segment deliberately, place monitoring that understands industrial protocols, and test the result before you rely on it.

Slide 15 · Secure Design

Chapter three is secure design. We look at infrastructure defined in code rather than by hand, at the network patterns that contain an intruder, and at how security controls are chosen in the first place. Design decisions last for years, so they are worth getting right. Get the design right and everything after it is easier. Get it wrong and no amount of patching will fully save you. A network built without thought about containment is a network where one stolen laptop becomes an estate wide incident.

Slide 16 · Infrastructure as Code

Infrastructure as code replaces manual configuration with machine readable files. Infrastructure as Code means the same file builds the same server every time, so environments cannot drift apart. Because those files live in version control, you keep a history of every change and can roll back to a known good state. Automated scanning catches a weak setting before it reaches production. The benefit is consistency, and consistency is where most security actually comes from.

Slide 17 · Infrastructure as Code Risks

But the same power creates a matching risk. A misconfiguration in one file is reproduced everywhere that file is used, so one mistake becomes an estate wide mistake. Secrets are the other trap, because keys and passwords written into configuration files leak through repositories. So use a dedicated secrets manager, never the file. Add peer review, run automated validation in the pipeline, and keep development, test, and production environments separate. Then audit and test the configurations as you would any other system.

Slide 18 · Secure Network Design

Secure network design begins with defence in depth, which means layers of control so that one failure is not fatal. Segmentation stops an intruder moving freely, and a virtual local area network is the usual way to do it. An access control list then states which segment may talk to which. Public facing services belong in a demilitarised zone, a buffer between the internet and the internal network. Least connectivity is the rule: allow only the connections the business truly needs.

Slide 19 · Protecting the Channel

Once data leaves a trusted segment, you must protect it in transit. A virtual private network encrypts traffic across an untrusted network. Transport Layer Security secures web traffic, and it begins with a handshake that authenticates the parties and agrees the keys. Internet Protocol Security works at the network layer and protects whole networks in tunnel mode. Administrators should reach systems through a jump server or bastion host, never directly. And certificates must be issued, renewed, and revoked by a trusted authority.

Slide 20 · Security Control Selection

Control selection is a decision, not a shopping list. Risk assessment comes first, because it tells you which threat deserves the budget. Compliance requirements then dictate some controls outright, such as encrypting cardholder data. Cost matters, including the ongoing cost of running the control, and so does operational impact, because a control that blocks the business will simply be bypassed. Controls are technical, administrative, or physical, and you want all three working together. Then test and monitor them through a Security Information and Event Management system.

Slide 21 · Data Protection

Chapter four is the data itself, because that is what all of this exists to protect. We look at the types of data that carry legal weight, the states data can be in, how classification drives handling, and the controls that protect data at rest and data in transit. You cannot defend what you have not identified, so identify the data and label it first. Type sets the legal obligation, and state sets the control you need.

Slide 22 · Data Types and States

Some data types carry legal weight. Personally identifiable information can identify a person, so its exposure enables identity theft. Protected health information is medical data covered by health privacy law. Financial data such as card numbers falls under the payment card industry standard. Then there are the three states. Data at rest sits on storage. Data in transit crosses a network. Data in use is being processed in memory. Each state has its own protection.

Slide 23 · Classification Levels

Classification tells you how much protection data deserves. Public data can be shared freely, such as marketing material. Internal data stays inside the organisation but is not sensitive in itself. Confidential data is sensitive, including personal information and customer records, and disclosure could cause legal or competitive harm. Restricted data needs the highest protection, such as trade secrets and classified material, with access limited to a need to know. You cannot protect what you have not labelled.

Slide 24 · Classification Drives Handling

Classification only earns its keep when it changes behaviour. Storage depends on the label, so restricted data must be encrypted with tight access control, while public data needs nothing special. Transmission follows the same logic, with confidential and restricted data encrypted and sent over trusted channels. Disposal matters too, because a drive sold with restricted data still on it is a breach. Label the data, physically or as metadata, so users know what they are holding. Then train people until classification becomes habit rather than paperwork.

Slide 25 · Protecting Data at Rest

For data at rest, encryption is the main control. Full disk encryption protects everything on the drive, which is what a stolen laptop needs, and BitLocker and FileVault are the familiar examples. File level encryption is more selective; the Windows Encrypting File System is the classic case. Databases can encrypt whole tablespaces or individual columns. All of it depends on key management: generate keys securely, store them safely, rotate them, and restrict who may use them. A hardware security module holds keys in tamper resistant hardware.

Slide 26 · Secure Disposal and Physical Protection

Disposal is where good intentions leak. Wiping overwrites the data so that it cannot be recovered, and it works on both hard disk drives and solid state drives. Degaussing uses a strong magnet, which destroys magnetic media but does nothing useful to solid state drives. Physical destruction by shredding, crushing, or incineration is the strongest option, and the right one for highly sensitive media. None of it is worth much if the server room door is unlocked, so pair encryption with physical access control and environmental protection.

Slide 27 · Protecting Data in Transit

For data in transit, never assume the network is safe. Use the secure version of every protocol you already know: Hypertext Transfer Protocol Secure for the web, Secure File Transfer Protocol for files, and Simple Mail Transfer Protocol Secure for mail. Email itself can be encrypted with Pretty Good Privacy. Data loss prevention watches outbound traffic and blocks sensitive content, such as card numbers, from leaving. Finally, validate certificates properly, because accepting a warning is exactly how an attacker in the middle succeeds.

Slide 28 · Resilience and Recovery

Chapter five is resilience, because security also means keeping the service alive. We look at high availability, alternate sites, backup and recovery objectives, power and environmental controls, and how all of it is tested. Resilience is a design habit, built from redundancy, planning, and rehearsal. A service that has never been failed over on purpose is a service nobody really understands. Ask what happens when each component fails, and design that answer before the failure.

Slide 29 · High Availability Concepts

Availability is measured as uptime, and by mean time between failures, which tells you how often a component gives up. Redundancy comes in layers: a redundant array of independent disks mirrors data across drives, network cards are teamed, and clustered servers take over from each other. A load balancer spreads traffic and checks health before sending work. Failover moves control to a standby system, and automatic failover beats a human with a phone. The aim throughout is to remove single points of failure.

Slide 30 · Alternate Sites and Geography

If an entire site is lost, you need somewhere else to run. A hot site is fully equipped with current data, so it can take over almost immediately, and it is the most expensive. A warm site has the hardware in place but needs data restored and systems configured before it can work. A cold site is an empty shell with power and cooling, which is cheap but slow. Geography is the balancing act: far enough apart that one flood cannot take both, close enough that replication stays acceptable.

Slide 31 · Backup Strategies and Recovery Objectives

Backups are the last line of defence, and the recovery objective drives the design. A full backup copies everything, so restoring is simple but slow. An incremental backup captures changes since the last backup of any kind, but restoring needs the full backup and every increment since. A differential backup captures everything since the last full backup, which restores faster than incremental. The Recovery Point Objective states how much data you can afford to lose. The Recovery Time Objective states how long you can be down.

Slide 32 · Power, Environment and Platform Diversity

Resilience also depends on the room the equipment sits in. An uninterruptible power supply carries the load through a short outage, and a generator covers a long one, so the two are used together. Cooling prevents overheating, fire suppression protects the hardware, and humidity control stops static discharge and condensation. Platform diversity is the strategic answer: use different operating systems, different applications, and more than one vendor. Then a single flaw, or one supplier failing, cannot take everything down at once. Diversity is cheap insurance.

Slide 33 · Testing Resilience and Recovery

An untested plan is only a guess. A tabletop exercise is a discussion, where the team walks through a scenario and finds the gaps in roles and communication without touching a system. Simulation testing actually runs the components, stressing systems and modelling failures. Full failover testing shuts the primary down and proves the backup really can carry the load. Parallel processing runs recovery alongside production, so you validate it without disruption. Document every test, because the lessons learned are the real output.

Slide 34 · Break

Ten minutes. When we return: business continuity planning, the exam map, and the quiz.

Slide 35 · Consolidation

Chapter six brings it together. We finish with business continuity planning, the umbrella that holds recovery, resilience, and the wider organisation obligations in one place. It is where a technical decision becomes a business commitment, and where a plan is either rehearsed or quietly abandoned. A plan nobody has read is

not a plan at all. Continuity is not a document that lives in a drawer; it is a set of rehearsed actions owned by named people. So let us look at how it is built.

Slide 36 · Business Continuity Planning

Business continuity planning is the umbrella over everything else. It starts with a business impact analysis, which identifies the critical processes, their dependencies, and the downtime each one can tolerate. Continuity of operations planning keeps those functions running from an alternative arrangement, and disaster recovery for IT is the part that restores systems and data. The plan is never finished. Update it when the business changes, and rehearse it with the people who will have to use it. Continuity is a business plan, with IT inside it.

Slide 37 · What this maps to in the exam

Quick alignment with the paper. Architecture models: on premises, the deployment models, and the service models with their shared responsibility. Compute and edge: hypervisor types, virtual machine escape, containers, serverless, the internet of things, and industrial control. Secure design: infrastructure as code, defence in depth, segmentation, and control selection. Data protection: types and states, classification, encryption at rest and in transit, and disposal. Resilience: availability, alternate sites, backup strategies, the recovery objectives, and continuity. Expect scenarios rather than definitions.

Slide 38 · Quiz

[Direct candidates to [Quiz_Session_07.html](#). 15 minutes.] Open the session quiz. Twenty questions, immediate feedback, and every question carries the domain and area it tests. Read every explanation, because the distractors are chosen to teach. Aim for seventy percent or better.

Slide 39 · Key takeaways

The takeaways to carry forward. Architecture sets where the trust sits, so know the shared responsibility line and secure the hypervisor. Virtualisation, containers, and serverless shrink what you manage, but not what you must secure. Devices and industrial systems need segmentation, monitoring, and lifecycle discipline. Infrastructure as code gives consistency, and also misconfiguration at scale. Classification decides handling, and labels make it real. Encrypt at rest and in transit, and manage the keys. And resilience comes from redundancy, recovery objectives, and plans you have actually tested.

Slide 40 · Revision checklist

Before next session: retake the quiz until you score seventy percent without guessing. Find the line, by picking one cloud service you use and writing down what you still own in it. Classify something real, by choosing a dataset at work and assigning it a level. Name your recovery objectives, the recovery point and recovery time for your most critical system. And read ahead to Session 8, network security controls and segmentation.

Slide 41 · Next session preview

Next time, Network Security Controls and Segmentation. We will cover physical security and deception technologies; audits, regulatory compliance and key management; segmentation for special environments, including guests and personal devices; and device hardening, network access control and filtering. Everything there stands on the architecture you now understand, and on the data you have classified. If you take one habit from today, make it this: classify the data, know where it lives, and design so that one failure is never the end. Thank you for your attention, and I will see you in Session Eight.